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COMPREHENSIVE REVIEW

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Opioid use, prescribing and fatal overdose patterns among racial/ethnic minorities in the United States: A scoping review and conceptual risk environment model

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Abstract

Issues: To date, there has been no synthesis of research addressing the scale and nuances of the opioid epidemic in racial/ethnic minority populations in the United States that considers the independent and joint impacts of dynamics such as structural disadvantage, provider bias, health literacy, cultural norms and various other risk factors.

Approach: Using the "risk environment" framework, we conducted a scoping review on PubMed, Embase and Google Scholar of peer-reviewed literature and governmental reports published between January 2000 and February 2024 on the nature and scale of opioid use, opioid prescribing patterns, and fatal overdoses among racial/ethnic minorities in the United States, while also examining macro, meso and individual-level risk factors.

Key Findings: Results from this review illuminate a growing, but fragmented, literature lacking standardisation in racial/ethnic classification and case reporting, specifically in regards to Indigenous and Asian subpopulations. This literature broadly illustrates racial/ethnic minorities' increasing nonmedical use of opioids, heightened burdens of fatal overdoses, specifically in relation to polydrug use and synthetic opioids, with notable elevations among Black/Latino subgroups, in addition uneven opioid prescribing patterns. Moreover, the literature implicates a variety of unique risk environments corresponding to dynamics such as residential segregation, provider bias, overpolicing, acculturative stress, patient distrust, and limited access to mental health care services and drug treatment resources, including medications for opioid use disorder.

Implications: There has been a lack of rigorous, targeted study on racial/ethnic minorities who use opioids, but evidence highlights burgeoning increases in usage, especially polydrug/synthetic opioid use, and disparities in prescriptions and fatal overdose risk-phenomena tied to multi-level forms of entrenched disenfranchisement.

Conclusion: There is a need for further research on the complex, overlapping risk environments of racial/ethnic minorities who use opioids, including deeper inclusion of Indigenous and Asian individuals, and efforts to generate greater methodological synergies in population classification and reporting guidelines.

KEYWORDS
harm reduction, medications for opioid use disorder, opioid use, racial disparities

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1 | INTRODUCTION

The opioid epidemic's slowly crystalising racial and ethnic dimensions punctuate the crisis' enduring intricacy and entrenchment in the United States [1, 2]. Current racialised patterns in opioid use initiation and overdose have largely been stoked by the introduction of fentanyl, a synthetic opioid, to drug markets as part of what has been termed the 'fourth wave' of the opioid epidemic. This fourth wave also speaks to the rise of polydrug use—for example, combining opioids with benzodiazepines and stimulants such as crack cocaine and methamphetamines [3, 4]. To date, there has yet to be a focused synthesis of the opioid epidemic in the United States along the lines of race/ethnicity that also considers existing and emergent factors linked to over/under prescibing and fatal overdose.

Prior reviews on the broader topic of racial variation in opioid use have focused on one racial/ethnic minority group, such as Black individuals [5–7], and have neglected deeper consideration of structural, meso and individual-level risks factors underlying increased use, prescribing and fatal overdose in these populations [8–12]. The wider effort to contextualise these populations is hampered by intricacies in racial (and ethnic) classification and under-sampling of minority groups in substance use research, both being dynamics associated with the field's underlying racial and epistemic biases [5, 13, 14]. Scoping reviews represent a structured approach to assessing the nature and dimensions of a body of literature when the theoretical and conceptual dimensions are unknown or amorphous [15].

The first aim of this article is to outline recent studies providing estimates of nonmedical and illicit opioid use (including opioid pain pills, heroin, fentanyl and fentanyl analogues), clinical opioid prescribing trends and fatal opioid-related overdose patterns among racial/ethnic minorities in the United States. The second aim is to broadly identify and characterise risk factors for opioid use/fatal overdose in these groups, focusing on structural, meso and individual-level factors.

2 | METHODS

Between 16 April 2022 and 19 February 2024, we conducted a scoping review to identify and characterise: (i) peer-reviewed research focused on the racial and ethnic correlates of nonmedical opioid use, opioid prescribing and fatal overdose; and (ii) risk factors for nonmedical opioid use/overdose among racial/ethnic minorities in the United States. This article utilises scoping review methods and approaches from Peters et al. [16], while the data charting and synthesis procedures were guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension for Scoping Reviews (PRISMA-ScR) [17]. The review is registered on Open Science.

2.1 | Inclusion and exclusion criteria

An article or report was included in the scoping review if it met the following inclusion criteria: (i) it appeared in a peer-reviewed journal or was a report from a government-based entity such as the Centers for Disease Control and Prevention; (ii) it provided survey-based estimates or measurable details on patterns of nonmedical opioid use or opioid use disorder (OUD), opioid prescribing or fatal overdose; (iii) there was subgroup analysis or contextualisation of outcomes of racial/ethnic minorities in the United States including Black/African American individuals, Latino/a/x individuals, Asian and Pacific Islander individuals and/or Native American/Indigenous individuals (i.e., differences in use, fatal overdose or opioid prescribing rates according to race/ethnicity); (iv) it contained data collected at some point between January 2000 and February 2024 (to provide a contextual trajectory into current patterns in the opioid epidemic, with an emphasis on literature addressing the fourth wave); (v) the data were collected on individuals in the United States; and (vi) it was written in the English language. Articles were excluded if the study was conducted outside of the United States or if there was no inclusion of non-White individuals. Moreover, we did not assess or include conference abstracts. There were no restrictions on study designs (e.g., cross-sectional vs. randomised, etc.) or sample sizes.

We located considerable variability in how variable-specific prevalences were defined and reported and how outcomes were generally measured (e.g., in terms of units of analysis, timeframes, etc.). Moreover, data quality and general reporting standards, namely in regards to OUD and (nature of) fatal overdose, vary from state-to-state [18, 19]. Additionally, we found differences in the racial (and/or ethnic) classifications that were used by authors. For example, some authors described members of their samples solely in terms of race (e.g., as 'Black' (sans ethnicity)), while others noted both the race and ethnicity of their sample participants (e.g., describing them as non-Hispanic Black). We deemed either characterisation to be acceptable. However, these inconsistencies, among others later highlighted, precluded complete racial aggregation, meta-analysis and computation of raw totals/percentages across the studies reviewed. Thus, and more broadly speaking, we decided to perform a scoping review of these elements, providing an overview, contextualisation

and critique of existing literature, noting these limitations, as appropriate.

2.2 | Search strategy

Articles were located and collated by two reviewers (Jerel M. Ezell and Babatunde P. Ajayi) using the keyword search parameters identified in Appendix 1. A third person (Ricky N. Bluthenthal) was brought in to resolve disputes. Manual electronic searches were initially performed on the Cochrane Library to determine if a similar review existed (it did not). Following this, we initiated structured, iterative searches on PubMed, Embase and Google Scholar. Search keywords focused on the epidemiologic surveillance of opioid use and OUD, prescribing and opioid use-related overdose (including via polydrug use and injection drug use) in racial/ethnic minority populations in the United States. Briefly, search terms included common opioid variants (e.g., heroin, hydrocodone, codeine, fentanyl, etc.), administration routes (e.g., injection, snorting, smoking, etc.), racial classifiers (e.g., Black, African American, Hispanic, Latino/a/x/ine, etc.) and drug use treatment/medications for opioid use disorder (MOUD) and harm reduction resources (e.g., buprenorphine, methadone, naloxone, fentanyl test strips, etc.). The reference sections of collated articles were also reviewed for potentially relevant content.

2.3 | Data charting and synthesis

PRISMA-ScR guidelines informed our charting and synthesis process [17]. Once the articles had been located, their references were loaded into Zotero, a citation management program. To ensure a degree of uniformity in charting approaches, a subset of five articles were charted by both primary reviewers, with consensus reached on the subsequent abstraction techniques. The articles' abstracts were read for relevant details, and then cues corresponding to article details were charted into a Microsoft Excel spreadsheet, using standard article trait parameters [20]. Specifically, for articles that were provisionally deemed eligible, the following details were captured: first author name, the year the article was published, the year that the data were collected and the study design (cross-sectional, randomised, etc.). During a second pass for the presence of racial/ethnic subgroup analysis, these additional details were captured: number of participants, the types of opioids/drugs used by the participants and the results of any stratification analyses done according to participants' race/ethnicity. A third scan captured the following details: (nonmedical) opioid usage rate, opioid-related overdose rate, opioid prescribing rate and potential risk factors for opioid use initiation/fatal overdose. Details were kept regarding reasons for including or excluding articles in the final review.

2.4 | Theoretical mapping

Following the provisional scoping exercise, we consulted with an external content expert and scoping review methodologist to refine the analytic approach [21]. Based on their feedback, we then abstracted information from the articles corresponding to Rhodes's risk environment model and its four domains, specifying physical, social, economic and policy-related risks [22]. In contextualising the risk environments that characterise the experience of racial/ethnic minorities who use drugs, a focus on the impact of structural impediments can be instructive [23, 24]. Key outcomes extracted from the articles focused on structural, meso and individual-level risk factors potentially corresponding to increased or decreased acceptability, access to and/or usage of social supports, harm reduction resources and drug use treatment. We also initially considered racial differences according to age, income, education and marital status, but these variables/intragroup differences were only infrequently reported in the literature's analyses. Therefore, we removed these measures from consideration in the scoping review, while acknowledging their import for future analyses in these domains. As part of our wider strategy, we executed an interpretive, risk environment-specific analysis of the extracted content. This comprehensive, theory-driven approach allowed us to better interpret and frame the intricate cross-cultural, hyper-spatial risk factors that animate this space.

3 | FINDINGS

Figure 1 outlines the article scoping review flow. A total of 1294 articles initially were located across PubMed ($n = 575$), Embase ($n = 377$) and Google Scholar ($n = 342$). From this initial batch, 164 duplicates were removed, bringing the cache to 1131 articles. Of the 1131 articles that had their titles and abstracts assessed for relevance, a total of 1037 were excluded for misalignment (i.e., they lacked direct or indirect connection to the research aims). This left 94 articles for full-text retrieval and further assessment for relevance. Of these, an additional 31 were removed because they: (i) lacked intergroup analysis of racial/ethnic differences in outcomes ($n = 11$); (ii) did not focus on individuals who use opioids (including polydrug use variations) ($n = 7$); (iii) represented non-governmental grey literature ($n = 7$); or (iv) were a poster/conference abstract ($n = 6$). In total, 63 of the initially reviewed batch of articles were found to meet the full

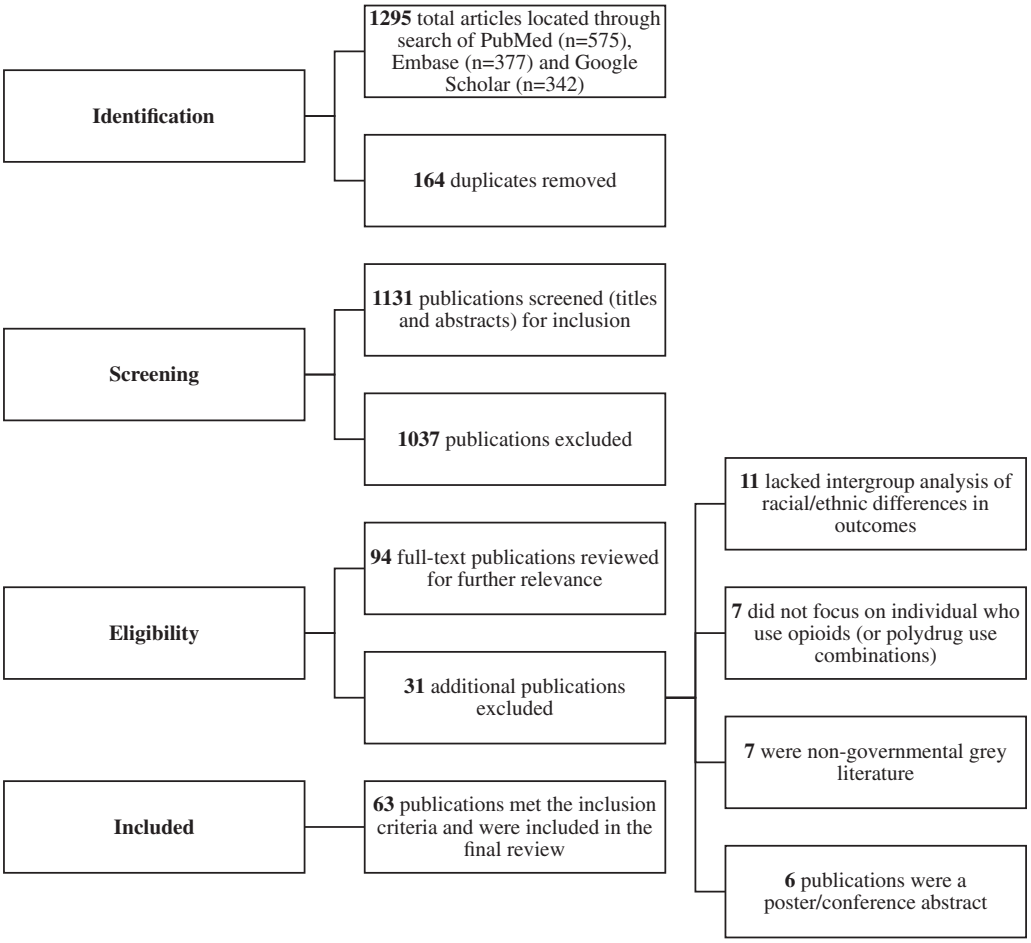

FIGURE 1 Scoping review flow diagram.

inclusion criteria and were included in the final scoping review and interpreted through Rhodes' risk environment model [22].

3.1 | Racial/ethnic patterns in opioid usage, prescribing and fatal overdoses

3.1.1 | Nature and scope of racial/ethnic disparities in opioid use

A number of studies assessed differences in opioid use according to race/ethnicity. In terms of study design, a large portion of these studies used cross-sectional and non-random samples, thereby limiting generalisability. Moreover, most of these studies used national or state-specific datasets, infrequently teasing out city or community-level nuances or differences and thus diminishing a richer understanding of meso-level dynamics. Table 1 outlines key findings related to the scope and scale of opioid use, opioid prescribing and overdose-related fatalities in racial/ethnic minority groups in the United States. The metrics used across these studies to report on these three outcomes were highly variable, foreclosing a uniform synthesis of findings.

Due presumably to differences in OUD diagnosis and reporting standards across states [28, 29], there has been limited research providing national-level OUD estimates according to race/ethnicity. Much of the existing research instead focuses on rates of 'nonmedical' opioid use; that is, usage of opioids substantially above and beyond clinical recommendations or use of any illicit opioid (e.g., heroin) of any volume, including in the context of polydrug use.

The literature broadly suggests that the current racial/ethnic hierarchy, from highest to lowest percentage of non-medical opioid use, is Indigenous individuals, followed by White, Black, Latino/x and Asian individuals. Although rates of prescription opioid misuse are generally lower among racial/ethnic minorities relative to White individuals, rates of use of "illicit" opioids, such as heroin, are frequently higher [30, 31]. These dynamics are magnified by the surge in synthetic opioid-involved overdoses, mostly involving fentanyl, among non-Hispanic Black, Latino and Indigenous individuals during the initial phases of

TABLE 1 Key racial/ethnic dimensions of the opioid epidemic.

Domain	Context and selected references
Nature and scope of disparities in opioid use	This refers to opioid usage in the United States from highest to lowest. Indigenous populations have the highest reports of nonmedical opioid usage, followed by White, Black, Latino and Asian populations, with rates rising most steadily among Black individuals during/post-COVID-19 [1–5].
Disparities in overdose-related fatalities	Recent research consistently illustrates an increasingly disproportionate burden of opioid-related deaths (mainly attributable to overdose) among Indigenous, Black and Latino individuals, relative to White individuals [2, 6–10].
'Illicit' opioid use, polydrug use and drug contamination	Heroin and fentanyl usage has been rising among Black, Indigenous and Latino populations relative to White individuals, this often being spurred by (unintentionally) mixing drugs with fentanyl and other synthetic opioids or stimulants such as crack cocaine [6, 11]. To this end, Black and Latino populations are more frequently exposed to contaminated drugs compared to White and other racial/ethnic minority populations [25–27].
Opioid (under) prescribing patterns	White individuals, relative to Black and Latino individuals, receive higher levels of pain management, including an increased volume of opioid prescriptions [12–17]. Much of this is likely driven by implicit and unconscious biases that lead clinicians to underestimate or ignore the pain experienced by racial/ethnic minority patients [18–22].

COVID-19 [32–36]. Nonetheless, even in consideration of pre-pandemic rates, heroin and (unexpected) fentanyl usage have generally been rising and have been higher among non-Hispanic Black, Latino and Indigenous individuals, relative to non-Hispanic Whites [37, 38].

Using data from the National Survey on Drug Use and Health from 2015 through 2018, Schuler et al. estimate the past-year prescription opioid misuse prevalence to be 6.3% in Indigenous populations, followed by 4.8% in non-Hispanic White populations, 3.9% in non-Hispanic Black populations, 3.8% among Latinos and 1.8% among Asians [39]. The authors further estimate heroin use at 0.4% among non-Hispanic White individuals and 0.2% among non-Hispanic Black individuals and Latinos. A study conducted on data from 2019 using the same data source found a closing gap between non-Hispanic Black and non-Hispanic White individuals for past-year prescription opioid misuse at 3.4% and 3.8%, respectively [40]. In a large, non-probability sample study of diverse individuals conducted in the United States in 2020, non-Hispanic White (4.81%) and non-Hispanic Black individuals (4.62%) had the highest lifetime "street" (illicit) opioid use followed by Hispanics (3.53%) and Asians (0.82%) [41]. However, in adjusted models, current use was higher for both street opioid use and nonmedical prescription opioid use among non-Hispanic Black and Hispanic individuals relative to White individuals. In terms of trends, using data again from the National Survey on Drug Use and Health for the years 2015 through 2019, prescription drug use among non-Hispanic White and Hispanic individuals decreased, while it remained the same for non-Hispanic Black and American Indian/Alaskan Native individuals [42]. Of note, across the literature, corresponding analyses for Indigenous and Asian individuals (overall or for subgroups) were rare, this likely due to—but not often articulated in the studies as limitations—small sample sizes for these populations and thus the analyses being underpowered to detect differences [43].

3.1.2 | Disparities in opioid prescribing and overdose-related fatalities

A substantive body of research has explored differences in patterns of opioid prescribing in clinical and pharmacy settings, frequently findings that non-Hispanic Whites, relative to racial/ethnic minorities, receive higher levels of pain management, including an increased volume of opioid prescriptions [44–63], emphasising a persistent trend.

The most robust existing data on racial/ethnic differences in opioid use pertain specifically to studies reporting fatal overdose rates; notably, these studies typically do not include estimates on rates of use. In addition to recent studies focused on within-state racial/ethnic differences in opioid-related overdose mortalities [14, 33, 64–72], a number highlight differences at the national level, largely relying on data from the National Vital Statistics System [38, 73–85]. Both state and national studies have typically concentrated on either annual overdose fatality rates or year-to-year percentage changes in fatalities (i.e., changes longitudinally across a multiple-year time period) in examining racial/ethnic patterns, typically locating disparities therein.

Using a national sample, Friedman et al. estimated prescription opioid-related deaths in 2020, finding elevated rates among Indigenous (41.4 per 100,000 people) and non-Hispanic Black individuals (36.8) as compared

to non-Hispanic White (31.6) and Hispanic individuals (17.3) [76]. In a pooled analysis of (any) opioid-related deaths from 2018 to 2019 in Kentucky, New York, Massachusetts and Ohio, researchers observed a 36% increase in opioid-related deaths among non-Hispanic Blacks and no increases in any other racial/ethnic group [68].

The race/ethnicity-focused literature on synthetic opioid-related deaths at the national level is limited. However, similar patterns appear to exist. For instance, Hoopsick et al. observed a heroin-related overdose death rate of 17.7 per 100,000 people for Black individuals versus 8.5 (each) for non-Hispanic White and Latino individuals [38]. Fentanyl-involved mortality rates for non-Hispanic White persons ranged from 0.7 to 0.8 per 100,000 from 2011 through 2013, then rose by 108.8% from 2013 to 2016 [84]. In contrast, non-Hispanic Black individuals had the largest annual percentage rate increase of fentanyl-involved deaths from 2011 through 2016 (140.6% per year), followed by Latinos (118.3% per year) [84]. These disparities sharpened during COVID-19. For example, in 2021, the highest age-adjusted rate for drug overdose deaths involved fentanyl and was among American Indian/Alaskan Native individuals at 33.1 per 100,000 people, 31.3 for non-Hispanic Black individuals, 24.6 for non-Hispanic White individuals, 14.1 for Hispanics and 2.3 for Asians.

3.2 | Risk factors for opioid use and associated morbidities in racial/ethnic minority populations

A variety of complex systemic and individual-level risk factors contribute to opioid use initiation and patterns of morbidity and mortality among racial/ethnic minorities in ways that both diverge and overlap with risk factors for White individuals (Table 2). Below, we outline some of the key elements in these domains, focusing on four risk environment forces [22]: (i) acculturative stress, as a cognate of mental illness; (ii) racialised policing and criminalisation of opioid use; (iii) spatial and geographic dynamics; and (iv) limited access to and trust in drug use treatment and harm reduction resources.

3.3 | The pronounced role of acculturative stress, mental illness and cultural trauma in drug use among racial/ethnic minorities

Acculturation relates to how minoritised groups are directly and indirectly integrated into a majority culture or aligned with the preferences of a powerful class, with pressures to adhere to specific cultural mores and behaviours to acquire and maintain social, economic and political capital [87, 88]. Along these lines, much of the mental illness experienced by racial/ethnic minority populations results partly from high levels of 'acculturative stress' [89]. Acculturative stress is the cumulative strain of working to meet the functionally untenable expectations of a socio-politically dominant (White) population and may intensify racial/ethnic minorities' feelings of isolation and 'otherness' [90–93].

Depression and anxiety are major risk factors for the initiation of drug use [94, 95], while also serving as frequent outcomes or exacerbated outcomes [96, 97]. The roughly 16% of Americans who have mental health disorders are responsible for roughly half of all opioids that are prescribed in the United States [98]. In the context of racial/ethnic minorities, opioids may be used to provide relief and distraction from recurrent experiences of racial oppression and disenfranchisement [99, 100], acting as a maladaptive form of coping [101]. In general, while Black, Latino and Indigenous individuals have lower overall rates of mental illnesses, they are underdiagnosed for conditions such as depression and anxiety [102, 103] and tend to have disproportionately high diagnoses (and overdiagnoses) of serious mental illnesses (schizophrenia, bipolar disorder, etc.) and elevated rates of culturally-specific psychological trauma [104, 105]. Indigenous and Black people may be especially susceptible to cultural trauma, specifically intergenerational trauma as brought on by their cultures' historic experiences with genocide, slavery and land dispossession at the hands of White colonialists and capitalists [106–108].

Latinos may be uniquely vulnerable to traumas derived from familial histories of forced migration (e.g., due to domestic crime and war, economic turmoil, etc. in their home countries) [109, 110]. Among racial/ethnic minority immigrants (broadly), deep, historic practices of large-scale resource and labor extraction coupled with present-day hyper-local domination by multinational corporations in their home countries may fuel opioid use by corroding wealth development prospects (as a protective factor) and positive cultural identity development [111]. Treatment programs that ignore or minimise the effects of these intricate forms of trauma may hence fail to adequately address the underlying causes of opioid use [112, 113].

Acculturative stress is a particularly potent risk factor for opioid use among youth, and first, second and '1.5' generation immigrant youth in particular [114–119], given the higher levels of social/emotional malleability and lower levels of resilience that characterise this age group. To this end, racial/ethnic minorities tend to have lower rates of help-seeking and treatment for mental illness in comparison to Whites [120, 121], in part due to intracultural stigma and insurance/geographic barriers, which can motivate non-medical use of opioids as a coping mechanism.

TABLE 2 Overview of key risk environments for nonmedical opioid use and associated morbidities among racial/ethnic minorities.

Risk factor	Context and selected references
Acculturative stress, mental illness and cultural trauma	This refers to the burden of sociocultural conformity that racial/ethnic minorities are tasked with while living in a society that caters to a White majority (or other powerful classes) and forges active discrimination (in health care, economic opportunity, housing, etc.). This creates a cycle of compounding stress as they (minorities) attempt to pursue social and economic progress [1]. Racial/ethnic minorities' mental illnesses, namely depression and anxiety, are often underdiagnosed or are stigmatised and downplayed by clinicians and other professional stakeholders [2]. These mental illnesses are broadly galvanised and amplified by cultural traumas associated with their ancestral exposure to racialised oppression such as colonisation, enslavement, resource dispossession, etc. This, along with lower levels of access to mental health care, lead to undertreatment [3, 4]. This susceptibility to mental illness and non-treatment or undertreatment results in nonmedical use of opioids as a coping mechanism [5–7].
Heightened racialised policing and criminalisation	The aftermath of the Reagan era's 'War on Drugs' has left racial/ethnic minorities, specifically Black and Latino individuals, relative to Whites, uniquely vulnerable to disproportionate levels of policing and imprisonment for even the most minor cases of drug use and distribution [8, 9]. Once in the carceral system, limited access to health care and MOUD limits racial/ethnic minorities' ability to recover or adjust [10, 11]. Following their release, community re-entry is often difficult due to the limited resources that populate their communities (as well as their new-found exclusion from employment, socialisation and housing opportunities); they become further isolated, increasing their chances of relapsing as well as social and sexual risk [12, 13]. Collectively, these dynamics may exacerbate their mental health problems, leading to initiation of opioid use or riskier modalities, such as injection drug use.
Disadvantaging spatial and structural dynamics	Owing to structural disinvestment, Black individuals in urban areas and the 'Black Belt' (i.e., Black communities in the [rural] south) often have levels of illicit opioid use, injection drug use and overdose-related morbidity and mortality that is higher than that of White individuals [14–20]. Latino populations may experience particularly acute forms of social isolation due to being disconnected from their primary culture and language, and Indigenous and immigrant communities may likewise experience this, in addition to intergenerational traumas wrought from colonialism and exposure to inadequate housing. The collective burden of racial discrimination, poverty, and lack of cultural connection may contribute to initiation of nonmedical opioid use [21–23].
Limited MOUD access and treatment engagement patterns	Black and Latino individuals have lower levels of access to MOUD, including both methadone and buprenorphine, largely due to the limited options typifying their communities [24, 28, 29]. This is reflective of the wide-reaching issue of health-care fragmentation and community disinvestment for racial/ethnic minority populations. These effects are further deepened by discrete disparities in MOUD treatment quality, engagement, and retention [32–34].
Disparities in naloxone access and utilisation	There are deficits in utilisation of naloxone among racial/ethnic minorities [35–37]. This is both a function of a lack of outlets in racial/ethnic minorities' communities to access these resources and lack of awareness of or faith in the safety or general utility of naloxone. Additionally, practitioners may be less likely to supply Black patients with naloxone (even after discharge for an overdose-related admission) and are more likely to view racial/ethnic minority patients with drug-related morbidities as 'drug-seekers', reinforcing tendencies to dismiss and undertreat this population [39, 44, 86].
Limited syringe services program, naloxone and product testing access/uptake	Limited research suggests that racial/ethnic minorities may be reluctant to engage with harm reduction sites and product testing. Black individuals are specifically less likely to access sterile injection equipment, naloxone, etc. from syringe services programs, both due to mistrust and a lack of accessible communal outlets [45–47]. Evidence on safe consumption sites acceptability by race/ethnicity is very limited and mixed [48, 49], while emergent evidence suggests Blacks may be less likely to use fentanyl test strips [50, 51]. The personal inhibitions of racial/ethnic minority individuals (in terms of their decisions to utilise certain resources like syringe services programs, fentanyl test strips, etc.) may reflect both a lack of awareness/education and a stark confrontation with structural barriers, both of which place them at a heightened risk of the adverse health effects that arise from using contaminated drugs or non-sterile injection equipment.

Abbreviation: MOUD, medications for opioid use disorder.

3.3.1 | Deep and persistent racialised policing and criminalisation

Because of the nation's War on Drugs [122, 123], Black and Latino individuals represent a highly socially vulnerable population that is disproportionately policed and imprisoned [124–126]. This heightened criminalisation is connected to the act of drug use in these groups as well as the manufacturing and distribution/selling of drugs in their communities [127, 128], punctuating a triadic, reproductive effect.

For racial/ethnic minorities who are imprisoned, rates of social isolation increase, and their access to general health services, including mental health care, becomes greatly limited [129, 130]. Moreover, MOUD access—and quality—in carceral settings is substantially diminished [131–133]. Collectively, these dynamics dramatically raise the likelihood of more hazardous forms of drug use and relapse and may also increase suicidality [134, 135]. Drug court graduation is also lower among non-White individuals despite non-White population size acting as a predictor of drug court involvement [136, 137]. Further, when implemented, low-level arrest rates and the likelihood of recidivism are higher for non-White than for White individuals [136, 138].

The period post-release is also associated with high risks of relapse [139–142]. Moreover, incarceration is a key risk factor in the development and worsening of mental illness, namely depression and anxiety [143, 144]. Given the limited social and communal resources that typify many racial/ethnic minority communities, reintegration is particularly difficult for diverse individuals [145–147], further heightening the likelihood of social network attenuation, communal disengagement, poor mental health, and usage of opioids and other drugs as a primary coping strategy.

3.3.2 | Residential segregation and spatial dynamics of opioid use among racial/ethnic minorities

Because it directly produces dynamics that encourage—or simply do not discourage—drug use and thereby curates specific spaces for opioid use, the built environment is highly associated with initiation, overdose and relapse [148]. With this in mind, much of the existing research examining the racial disproportionate contours of opioid use has focused on racial/ethnic minorities in urban environments [3, 149] and has primarily focused on heroin and/or polydrug use with crack [25, 31, 150] rather than prescription opioid misuse. Comparatively speaking, very little is known about non-urban racial/ethnic minorities who use opioids. Factors such as systemic racism, compounded by the limited medical and mental health resources and social supports that characterise low-income, segregated communities like those in urban areas, may drive and sustain disproportionately high-risk opioid use [151] in non-urban racial/ethnic minority populations [152]. Between 2013 and 2017, the steepest increases in opioid-related overdoses were highest among younger urban Black individuals (178%) followed by younger rural Black individuals (98%) [3].

Community disinvestment, driven by White flight and depopulation, contributes to more space for the marketing, distribution and usage of drugs [153–155]. To this end, Black individuals, relative to non-Black races, buy and use drugs in locations twice as close to where they live and may be farther away from syringe services programs, particularly in rural areas where drug use/harm reduction stigma may be comparatively higher, thus amplifying the likelihood of injection equipment sharing [156].

Similarly, rural Latinos, namely those living in migrant/farmworker communities, and Indigenous peoples, many of whom live on rural reservations, may experience elevated opioid use-related risks corresponding to the heightened social stressors and limited resources typifying their environments. Emergent evidence suggests that rural Latinos' physically intensive labour on farms and in other forms of seasonal work may spur usage—and eventual misuse—of prescription opioids and fuel illicit opioid use [119, 157, 158]. Likewise, communal-living Indigenous peoples' comparatively high levels of social isolation, limited economic opportunity and exposure to poor and inadequate housing (when living on reservations) may propel their nonmedical usage of opioids [159, 160].

3.3.3 | Lower access to and trust in drug use treatment and harm reduction resources

In general, individuals in segregated, rural and less densely populated communities have substantially lower levels of access to (high quality) drug use treatment, specifically MOUD such as buprenorphine [161, 162]. Compared to White individuals, Black individuals are less likely to access MOUD [163–168]. Forces contributing to racial disparities in access/uptake include structural and geographic barriers (e.g., health insurance coverage, availability of licensed providers, etc.) as well as misgivings about treatment purpose and efficacy [169], factors tied to health literacy and negative experiences with the health-care system that sow deep distrust [170–172].

A study of racial/ethnic segregation found that counties with highly segregated Black and Latino populations had more facilities to provide methadone (per capita), while counties with highly segregated White communities had more facilities providing buprenorphine, which is generally regarded as more efficacious, per capita [173]. Moreover, syringe service programs (and safe consumption sites) are effectively banned or forced to operate discreetly in multiple racially diverse parts of the United States, though particularly in the

rural 'Black Belt' (areas like Alabama in the southern United States) with large Black populations [174].

Like MOUD, naloxone, a vital tool in efforts to reduce mortality associated with nonmedical opioid use, has been demonstrated to be underutilised in Black populations [175–181]. Limited naloxone usage is partly due to deficits in access/availability of naloxone in these communities, as brought on by health-care fragmentation and socio-political resistance [179, 180, 182–184]. Further, limited naloxone usage is tied to limited overdose recognition knowledge, distrust of the tool and inadequate training on proper administration among both first responders and people who use opioids and members of their networks [175, 185]. Work further shows that clinicians are less likely to supply Black patients with naloxone, even after discharge for an overdose-related admission, [186, 187] with data also showing emergency department clinicians are more likely to view Black patients with drug-related morbidities as 'drug-seekers' [24, 188, 189] and thus undeserving of (quality) care, further illumining the importance of understanding (medical) risk environments.

4 | DISCUSSION

Results from this scoping review and conceptual risk environment analysis highlight the complex socio-racial constellation of the opioid epidemic and the orbit of risk factors that have contributed to and sustain its increasingly disproportionate impacts. Literature assessed here reveals several key findings. First, there is a growing burden of opioid use, including of prescription and 'illicit' varieties, and of overdose-related mortality in racial/ethnic minority populations, particularly Black and Indigenous populations. Our review signals that most of the recent and current literature that possesses racial/ethnic subgroup analyses provides estimates on *nonmedical* opioid use only, rather than OUD. Nonmedical opioid use is not an immediate diagnostic proxy for OUD [18], a reality that takes on added gravity given the ways in which health conditions are frequently under or overdiagnosed among racial/ethnic minorities [113, 190]. This diagnostic conundrum casts into sharp relief the intricacy inherent to the process of gauging opioid use-related risks, whether in terms of understanding pathways to initiation, overdose or relapse, and in then extrapolating to appropriate primary or secondary interventions. Additionally, the literature suggests that Black individuals may be underprescribed opioids [58]. These tendencies, attached in equal parts to structural disadvantage and provider bias [191], can contribute to patients seeking illicit opioids as a means of pain relief or coping [192–194].

In terms of overdose patterns, our review located a number of articles suggestive of a broadly growing and unequal burden of opioid-related overdoses among racial/ethnic minorities, namely Black individuals. Of the studies examining trends in fatal overdose, a substantial number pointed to the outsized role of fentanyl and stimulants as part of individuals' polydrug use behaviours (i.e., the fourth wave of the opioid epidemic). Evidence indicates that rates of polydrug use, namely the use of opioids in conjunction with stimulants such as crack cocaine and methamphetamine, are accelerating among Black and Latino individuals [70, 195, 196]. However, it is presently unclear to what extent the polydrug use is intentional (i.e., the individual is knowingly mixing the drugs) or unintentional (i.e., the individual is not knowingly mixing the drugs but rather the drug supply was 'spiked' or contaminated by the manufacturer or dealer) [197, 198]. With this in mind, the widening body of research on this topic suggests that Black and Latino individuals have higher levels of exposure to contaminated drugs [25–27].

Utilising risk environment theory [188], we sought to apply the findings from the review to identify and contextualise structural, meso and individual-level risk factors among racial/ethnic minorities that might drive opioid use initiation, relapse or overdose. 'Upstream' factors that emerged included acculturative stress, racialised policing and criminalisation, and spatial segregation, all of which are directly tied to racial capitalism [199]. In addition to a wholesale rejection of the War on Drugs that racial capitalism is heavily oxygenated by [123], there is a need for structural interventions focused on reversing racialised community disinvestment, addressing health-care fragmentation, and providing trainings to reduce provider bias. Interventions such as these will help to tend to the acculturative stress and related sequalae that motivate individuals to consume opioids.

At the individual-level, deficits in health literacy (e.g., understanding risks associated with opioid use, knowledge of how to reverse an overdose, etc.) and distrust in/limited access to MOUD, naloxone and other resources, 'downstream factors' in the racial capitalism continuum, also constitute vital intervention targets. Owing to a legacy of medical disenfranchisement that dates back to the colonisation of the Americas by Europeans and enslavement of Africans [200], there is steep distrust of the health-care system and cultural animus around the joint processes of social and economic marginalisation and criminalisation that have been stoked [201]. Providers and advocates can more directly address relational chasms with racial/ethnic minorities through targeted outreach, education and engagement efforts, as well as acknowledgment of this stark historical context [202–205].

4.1 | Limitations

In general, the literature on several domains under consideration in this review—including prevalence of opioid use among racial/ethnic minorities and prescribing and overdose dynamics among Indigenous and Asian individuals—was limited, thus preventing a more holistic view of the contours of the opioid epidemic among racial/ethnic minorities *writ large*. Second, along these lines, there was considerable variation in how (and indeed whether) race and ethnicity were characterised and reported. This made a holistic synthesis of the opioid use-related traits and outcomes of all racial/ethnic minority groups in the United States ultimately untenable. To this end, there was also substantial variability in how opioid types/subtypes, combinations used and associated morbidities were described and also in time periods of reporting. Finally, epidemiologic surveillance approaches and reporting standards for fatal and nonfatal overdose vary considerably [28, 206], and thus it is possible that there were over or undercounts in the prevalence data reviewed here. These cleavages converged to preclude the possibility of a meta-analysis.

Finally, we note that our review did not specifically focus on several key correlates of overdose and OUD, namely gender, age and sexual orientation. Data show that rates of opioid-related overdose may be particularly high among transgender/non-gender-conforming non-Hispanic Blacks and Latinos [207] and among Indigenous and Latino youth [159, 208, 209]. Most studies included in the analysis did not have a rigorous focus on these intersectional dimensions [210, 211], and thus it was not possible to conclude the extent to which other key demographic factors, alone or in conjunction with race/ethnicity, may have influenced the outcomes of interest.

5 | CONCLUSIONS

Findings from this scoping review demonstrate wide and deepening racial and ethnic inequities in multiple axes of opioid use, including multiple minority groups having an elevated burden of fatal overdoses (specifically in relation to synthetic opioids). Complex risk environments that involve structural elements like community disinvestment, provider bias, and limited access to mental health and drug use treatment and harm reduction resources are centrally implicated. Furthermore, this review points to a bevy of methodological gaps in terms of both the breadth and quality of research on racial/ethnic minorities who use opioids, with particularly limited research on Indigenous and Asian individuals. These findings illuminate the need for more rigorous, longitudinal mixed methods research on racial/ethnic minorities *writ large* as a means of augmenting the cultural responsiveness of primary and secondary interventions. With this in mind, there is also an adjoining need to develop, implement, and evaluate culturally adapted harm reduction and drug use treatment interventions to create potentially effective means of support and health promotion for these uniquely vulnerable populations.

AUTHOR COTRIBUTIONS

Jerel M. Ezell was responsible for the conception of the study, locating articles, and writing and editing the manuscript. Mai Pho was responsible for locating articles and editing the manuscript. Babatunde P. Ajayi was responsible for locating articles and editing the manuscript. Elinor Simek was responsible for locating articles and editing the manuscript. Netra Shetty was responsible for editing the manuscript. Dawn A. Goddard-Eckrich was responsible for locating articles and editing the manuscript. Ricky N. Bluthenthal was responsible for the conception of the study, locating articles and editing the manuscript.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest, financial or otherwise, to report.

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APPENDIX 1

SEARCH TERMS UTILISED IN THE SCOPING REVIEW

We used a variety of search terms, and permuted terms, in an effort to capture the broad spectrum of peer-reviewed articles corresponding to the research aims guiding this scoping review. The search terms included: Abuse; addiction; amphetamine; benzodiazepines; binge; buprenorphine; cocaine; crack; decedent; death; drug; drug use; ecstasy; fentanyl; fentanyl testing strips; geography; harm reduction; heroin; ice; injection; injection drug use; ketamine; kratom; local; LSD; mescaline; methadone; methamphetamines; molly; naloxone; Narcan; national; needles; needle exchange; needle exchange program; nonmedical drug use; opioids; opioid use; opioid use disorder; overdose; overdose prevention center; painkillers; PCP; peyote; polydrug use; prescribing; prescription; prescription monitoring program; product testing; relapse; safe consumption sites; state; stimulants; synthetic; syringes; syringe exchange program; syringe services program; substance use; substance use disorder; and surveillance.

Permuted terms meant to retrieve articles that focused specifically on racial/ethnic dimensions of opioid use included: African American; American Indian; Alaskan; Asian; Asian American; Black; Caucasian; Hawaiian; Hispanic; Indian; Indigenous; Latino/a/x/ine; Native American; Pacific Islander; and White. We also used specific nationalities for Hispanic groups that are prevalent in the US, including the terms Cuban; Dominican; Mexican; and Puerto Rican. In this article, we use the expression 'Indigenous' to refer to Alaskan Natives, American Indians and Native Americans, and use the term 'Asian' to refer to Asian Americans and Pacific Islanders.